

CHANFORMER: An Encoder-Only Channel-Wise Transformer for Long-Term Multi-Asset Financial Time Series Forecasting

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ABSTRACT

Forecasting financial price series encounters significant obstacles due to intense volatility, sudden changes in distributions, and intricate inter-asset correlations that vary with market conditions, which intra-sample multivariate approaches cannot properly handle. This study introduces CHANFORMER, a novel encoder-only channel-oriented Transformer designed for long-term prediction over closely linked cryptocurrencies and stocks. Through incorporating reversible instance normalization to address the previous bottlenecks, CHANFORMER compresses each channel's full input history into one high-dimensional token via projection, utilizes a standard Transformer encoder for pure cross-channel self-attention to capture interdependencies, and incorporates a dedicated channel-mixing multi-layer perceptron to enable dynamic non-linear fusion across assets. CHANFORMER concurrently produces predictions for 50 OHLCV channels per cryptocurrency and stock group. Empirically, CHANFORMER surpasses state-of-the-art Transformers with average error decreases of 29% relative to closest rivals and 83–94% against classical variants. Throughout backtesting strategies, the long-short approach delivers robust positive total returns of 59% in crypto with Sharpe 1.09 over 364 days, and the architecture ensures operational viability with approximately 23 minutes of training along with 1.6 seconds inference time at a total of 0.107 kWh and 0.051 kg CO₂ computation overhead. Therefore, this advances channel-centric designs for high-dimensional multi-asset analysis in financial sequence prediction. Source code is publicly available at: github.com/TheQuantScientist/ChanFormer

1. Introduction

Cryptocurrency price sequences exhibit unique hurdles due to their inherently volatile and non-stationary nature, driven by a complex web of interconnected yet fundamentally erratic influences, such as market sentiment, regulatory shifts, and macroeconomic events (Ştefan Cristian Gherghina and Constantinescu, 2025; Chen, 2025). These characteristics set crypto assets apart from time series in more predictable fields like meteorology or manufacturing. Investors often manage multifaceted portfolios that traverse diverse market conditions, which intensifies the role of inter-asset dynamics as effects that vary across regimes are notoriously hard to predict. As a result, even cutting-edge time-series forecasting models deliver suboptimal performance in trading applications, primarily because they fail to account for these entangled relationships and context-specific patterns (Du, Cao, Jiang, Wang, Xu, Liu, Chen and Yuan, 2025).

This shortfall has motivated two parallel lines of inquiry: (1) deploying advanced neural architectures for multivariate prediction, capitalizing on their prowess in extracting hidden structures from feature-engineered inputs, interconnected series, and aggregated ensembles; and (2) rigorous analyses of relational networks among cryptocurrencies to bolster pattern recognition and model robustness. Transformer models, in particular, excel at encoding extended temporal dependencies through mechanisms like multi-head self-attention (Nie, Nguyen, Sinthong and

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Kalagnanam, 2022; Wu, Xu, Wang and Long, 2021; Zhou, Zhang, Peng, Zhang, Li, Xiong and Zhang, 2020), yet their tailoring to cryptocurrency finance remains underexplored compared to broader time-series datasets. On the other hand, hybrid approaches blending linear components offer enhanced transparency and stability against data drifts but struggle to encapsulate the nuanced interplay between assets in volatile, non-stationary environments (Mettu and Kandoth, 2025). Furthermore, while existing forecasting research frequently prioritizes isolated univariate problems or basic multivariate configurations, it tends to overlook the temporal sequences and interlinked motifs spanning multiple assets. In addition to within-series autocorrelations, the modeling process must integrate cross-asset linkages, especially among tightly coupled financial instruments, to augment real-time predictions. These linkages, however, add substantial intricacy, involving adaptive correlations that shift with market phases, risk-spreading advantages from noise diversification, and common underlying factors that transform the task into a high-dimensional puzzle. Tackling such interwoven complexities necessitates innovative, resource-demanding computational paradigms.

To address the pronounced non-stationarity, regime shifts, and strong cross-asset dependencies characteristic of cryptocurrency markets, we propose CHANFORMER, a functionally lightweight encoder-only Transformer tailored for multivariate, multi-asset, single-step forecasting. Instead of applying conventional patch-based tokenization along the time dimension (Nie et al., 2022), CHANFORMER flattens each asset's input sequence of length into a single high-dimensional token per channel, projecting the entire history and processing these independent tokens through a standard encoder. This design choice deliberately removes intra-asset temporal self-attention, transferring the modeling of long-range dependencies to the expressive power of the per-asset projection and encoder layers. Cross-asset interactions in diversified portfolio are explicitly captured via a dedicated channel-mixing multi-layer perceptron (MLP) applied after the Transformer, enabling dynamic non-linear information fusion at each prediction step. Reversible Instance Normalization (RevIN) is applied at both input and output to stabilize training under severe distributional shifts. The resulting CHANFORMER offers a favorable trade-off between computational efficiency, training stability, and the ability to learn regime-adaptive inter-asset relationships, delivering consistently stronger predictive performance than SOTA Transformers across long-term cryptocurrency and stock landscapes while generating decent backtesting profitability.

The paper is outlined as follows: Section 2 provides an overview of existing studies in financial time series modeling. Section 3 defines the research objectives and describes the proposed CHANFORMER architecture, including the evaluation metrics and important implementation settings. Section 4 presents the empirical findings, covering forecasting accuracy, computational costs, backtesting outcomes, and additional analyses. Section 5 interprets the key results and outlines directions for future work, while Section 6 summarizes the overall conclusions of the study.

2. Related Work

2.1. Statistical Architectures

Fundamentally, statistical time series models parameterize the joint probability distribution of a temporally ordered sequence by specifying how the conditional expectation and higher moments evolve as functions of lagged values, lagged disturbances, and possibly time-varying parameters or unobserved components (Gupta and Jaiswal, 2024; Miller, Aldosari, Saeed, Barna, Rana, Arpinar and Liu, 2024a). This recursive structure exploits the serial dependence inherent in temporal data to produce coherent multi-step-ahead forecasts, decompose trends and cycles, and quantify uncertainty under realistic violations of independence and homoskedasticity assumptions prevalent in real-world economic and environmental series. Their applications have been widely applied in predictive finance:

Using daily stock data from Apelo Pharmaceutical spanning 2020 to 2024, Li et al. (Li, 2025) employed an ARIMA-GARCH model and reported strong short-term forecasting performance, with a mean absolute percentage error (MAPE) of 3.81% and a mean squared error (MSE) of 0.438. These error metrics compare favorably to established benchmarks in financial time series forecasting. The hybrid model effectively captured the conditional mean through ARIMA's differencing combined with autoregressive and moving average components, while GARCH accounted for the observed time-varying volatility in the residuals. Similarly, a composite ARIMA-GARCH specification applied to daily closing prices of Amazon (2020–2022) and Caterpillar (2021–2022) produced lower root mean squared error (RMSE) and MAPE values than a standalone ARIMA model, resulting in an approximate 25% improvement in short-term predictive accuracy (Wang, 2024). Comparable gains were documented for Apple stock returns, where the ARIMA-GARCH approach reduced the mean absolute error (MAE) to approximately 1.15%, RMSE to 1.32%, and MSE to 0.02 relative to simpler benchmarks (Xu, 2025). In an analysis of Coca-Cola monthly prices from October 2024 to April 2025, Qi (Qi, 2025) found that an ARIMA(0,2,1) model achieved an out-of-sample MAPE of 2.93% for short-term forecasts. Diagnostic testing indicated weak ARCH effects in the residuals, rendering additional volatility

modeling via GARCH unnecessary. For Indonesian banking stocks over the period 2014–2018, ARIMA outperformed GARCH in terms of RMSE, MAE, and MAPE in three of the five years examined, whereas GARCH delivered good performance in the remaining two years (Nuryatin, 2020). This pattern underscores that the relative effectiveness of conditional mean modeling (ARIMA) versus explicit volatility clustering modeling (GARCH) can depend markedly on the specific time period and prevailing market conditions.

In the context of cryptocurrency markets, an ARMA-GARCH model fitted to Bitcoin returns yielded an MAE of approximately 0.024 and an RMSE of 0.037, corresponding to an RMSE reduction of roughly 11.27% compared with prior ARMA-GARCH benchmarks following careful parameter selection (Anand and Sharan, 2025). The ARMA component addressed linear dependencies in the return series, while GARCH accommodated the pronounced conditional heteroskedasticity typical of cryptocurrency price behavior. Finally, Kareem and Aue (Kareem and Aue, 2025) demonstrated that the exponential GARCH (EGARCH) specification outperformed both ARIMA and seasonal ARIMA (SARIMA) in modeling Bitcoin volatility. The dominance of EGARCH was attributed primarily to its ability to capture the asymmetric impact of positive and negative shocks, a salient characteristic of cryptocurrency price dynamics. Overall, GARCH and ARIMA-based architectures are highly trusted for financial prediction over Prophet or Regression variants (Keluli and Mauritsius, 2025; Raissa, 2025).

However, while the aforementioned statistical models have demonstrated robust performance in capturing linear dependencies and volatility clustering in relatively stable time series, they increasingly fall short in extreme non-stationarity, abrupt regime shifts, and multifaceted nonlinear interactions prevalent in multivariate cryptocurrency and high-frequency finance data of diversified portfolios (Hall and Rasheed, 2025). Accordingly, these probabilistic architectures rely on stringent assumptions of homoskedasticity or symmetric shock responses that are routinely violated in crypto markets, where fat-tailed distributions, sentiment-driven anomalies, and exogenous shocks from global events can render parametric forecasts brittle and unreliable, often leading to substantial underestimation of tail risks and predictive inaccuracies during turbulent periods (Tjøstheim, 2025). This rigidity underscores the imperative for more adaptive, data-driven paradigms like deep learning, which can autonomously extract hierarchical features from vast, unstructured datasets to achieve stable generalization and resilience in real-world forecasting environments.

2.2. Deep Learning Architectures

Deep learning (DL) has long been successfully applied across diverse domains over statistical base, yet its advantages in time series forecasting emerged only gradually, as evidenced by the Makridakis (M) Competitions where hybrids blending statistical and machine learning techniques dominated early results, while pure DL lagged behind. Recent architectural breakthroughs such as attention-based encoder-decoders, Transformers, and graph neural networks (GNNs) have propelled deep learning to demonstrate substantial applicability in high-dimensional time series tasks (Miller, Aldosari, Saeed, Barna, Rana, Arpinar and Liu, 2024b; Anh and Ha, 2024).

Concerning stock forecasting, Anh and Hy (Anh and Hy, 2025) developed CLAM, a hybrid CNN-LSTM-AttentionMechanism model that outperforms standalone variants on healthcare stocks (e.g., ABBV, JNJ, C) in one-week-ahead forecasting, improving MAE and RMSE up to 80% while capturing 70% out-of-sample trends. Seabe, Moutsinga and Pindza (2023) evaluated three recurrent deep learning architectures on major cryptocurrencies and reported that the bidirectional long short-term memory (BiLSTM) model delivered the strongest performance, attaining mean absolute percentage errors of approximately 0.036 for Bitcoin, 0.041 for Litecoin, and 0.124 for Ethereum, along with lower root mean squared error compared to standard LSTM and GRU variants. The bidirectional structure enabled more effective capture of sequential temporal dependencies by processing historical price series in both forward and backward directions, thereby improving predictive accuracy over unidirectional counterparts. A separate study applied LSTM networks to forecast prices of AMP, Ethereum, EOS, and XRP over a 180-day horizon and obtained high regression coefficients of approximately 0.96 in both training and testing phases (Ammer and Aldhyani, 2022). The gate-based memory mechanisms of LSTM facilitated reliable learning of complex, non-linear patterns inherent in cryptocurrency time series. Biki, Sakamoto, Takei, Alam, Riajuliislam and Showaibuzzaman (2025)'s survey of deep learning approaches for cryptocurrency price prediction (Biki et al., 2025) concluded that hybrid LSTM-GRU models and standalone GRU architectures frequently outperformed basic LSTM configurations in terms of mean absolute percentage error and root mean squared error when applied to Bitcoin, Ethereum, and Litecoin series. These findings underscore the advantages of advanced gating mechanisms in modeling long-range dependencies within volatile financial data. Dashtaki et al. (Dashtaki, Dashtaki, Chagahi, Moshiri and Piran, 2024) integrated numerical price features with social media sentiment extracted via BERT into a bidirectional LSTM framework, resulting in approximately 96.8% accuracy in predicting the directional movement of cryptocurrency prices. This hybrid approach

demonstrated the substantial predictive benefit of combining structured time series inputs with unstructured textual sentiment signals. Other studies follow the same pathway of hybrid traditional modeling (Anh, Ha and Thai, 2024; Nguyen, Nguyen, Duong and Ha, 2026).

Recent Transformer-based models (Nie et al., 2022; Wu et al., 2021; Zhou et al., 2020; Zhou, Ma, Wen, Wang, Sun and Jin, 2022; Liu, Hu, Zhang, Wu, Wang, Ma and Long, 2023) have demonstrated state-of-the-art performance in multivariate time series forecasting. They leverage techniques such as decomposition with auto-correlation mechanisms, probabilistic sparse self-attention combined with distillation, and subseries-level patching with channel-wise independent processing across benchmark datasets including Weather, Electricity, and ETTh. Nevertheless, Transformer-based frameworks are highly potential. Across global stock regimes, LiteFormer, an encoder-only Transformer, achieved an average reduction of 3.45%–9.09% over vanilla Transformers and up to 48% over recurrent neural network models, producing instant inference at 38ms (Anh, Phat, Son, Nhan, Phien, Tuan and Kim, 2026a). Despite the growing success of Transformer architectures across various domains, their application to cryptocurrency price prediction is still relatively limited. Moreover, prior research has predominantly concentrated on ensemble techniques and feature engineering for multivariate forecasting of individual assets. This narrow focus neglects the potential benefits of multi-asset modeling, where different assets can be treated as interconnected features to capture cross-asset relationships and interdependencies. These motivate the research of the proposed ChanFormer.

3. Methodology

3.1. Problem Formulation

We address one-step-ahead forecasting of a panel of C financial time series comprising daily OHLCV observations from multiple assets. Let $\mathbf{X}_t = [X_{t,1}, \dots, X_{t,C}]^\top \in \mathbb{R}^C$ be the observation vector at day t across C channels ($C = 50$ for 10 crypto/stock assets). Given a lookback window of length L , the input at time t is

$$\mathbf{X}_{t-L+1:t} \in \mathbb{R}^{L \times C}. \quad (1)$$

The objective is to learn a mapping $f_\theta : \mathbb{R}^{L \times C} \rightarrow \mathbb{R}^C$ such that the forecast $\hat{\mathbf{y}}_t = f_\theta(\mathbf{X}_{t-L+1:t})$ approximates the true next-day vector $\mathbf{y}_t = [X_{t+1,1}, \dots, X_{t+1,C}]^\top$. The model is optimized via mean squared error:

$$\mathcal{L}(\theta) = \frac{1}{T} \sum_{t=1}^T \|\mathbf{y}_t - f_\theta(\mathbf{X}_{t-L+1:t})\|_2^2. \quad (2)$$

This task is particularly demanding in financial markets due to (i) pronounced non-stationarity and abrupt distributional shifts driven by volatility clustering, regime switches and news events, (ii) rich cross-asset dependencies induced by macroeconomic factors, sector correlations and contagion effects, and (iii) the necessity to jointly model dozens of interrelated yet heterogeneous channels while maintaining computational tractability and per-asset predictive accuracy. Purely univariate approaches discard valuable inter-market signals, whereas conventional multivariate Transformers often struggle with input non-stationarity and inefficient representation of long lookback windows. These challenges motivate ChanFormer, which prioritizes maximal cross-channel interaction and distribution-shift robustness over fine-grained temporal tokenization.

3.2. Proposed Architecture

We introduce ChanFormer, a channel-centric Transformer architecture designed for multi-asset one-step-ahead forecasting without temporal patching. The model processes the full historical sequence per channel as a single high-dimensional token, enabling direct cross-channel interaction while preserving long lookback information through linear projection rather than patch-wise tokenization. The key notations are summarized in Table 1.

The input $\mathbf{X}_t$ is first optionally normalized using Reversible Instance Normalization (RevIN). Theoretically, RevIN adaptively removes channel-wise non-stationarity (shifting means and volatilities) from the input via learnable affine parameters, thereby stabilizing gradient flow and enabling the downstream deep architecture to focus on learning genuine temporal and cross-asset dependency patterns rather than spurious distributional artifacts, while its invertible nature guarantees exact reconstruction of the original scale statistics at prediction time, preserving forecasting calibration under severe regime shifts (Kim, Kim, Tae, Park, Choi and Choo, 2022). Equations are as follows:

Table 1

Key notations for ChanFormer architecture.

Symbol	Description
$\mathbf{X}_t \in \mathbb{R}^{L \times C}$	Input matrix at time t
L	Lookback sequence length ($L = 90$)
C	Number of input channels ($C = 50$ or $C = 100$)
$\mu, \sigma \in \mathbb{R}^C$	Per-channel mean and standard deviation
ϵ	Numerical stability constant ($\epsilon = 10^{-5}$)
$\mathbf{w}, \mathbf{b} \in \mathbb{R}^C$	Learnable RevIN affine parameters
$\hat{\mathbf{X}}_t \in \mathbb{R}^{L \times C}$	Normalized input
d	Model dimension ($d = 384$)
$\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d \times L}$	Sequence-to-token projection matrix
$\mathbf{b}_{\text{proj}} \in \mathbb{R}^d$	Projection bias
$\mathbf{E}_{\text{pos}} \in \mathbb{R}^{1 \times 1 \times d}$	Learnable positional embedding (single token)
H	Number of attention heads ($H = 6$)
Λ	Number of Transformer encoder layers ($\Lambda = 4$)
$\mathbf{v}_c \in \mathbb{R}^d$	Final representation for channel c
$\hat{y}_c \in \mathbb{R}$	Normalized scalar prediction for channel c
$\tilde{y}_c \in \mathbb{R}$	Denormalized prediction

$$\mu_c = L^{-1} \sum_{i=1}^L X_{t,i,c}, \quad (3)$$

$$\sigma_c = \sqrt{L^{-1} \sum_{i=1}^L (X_{t,i,c} - \mu_c)^2 + \epsilon}, \quad (4)$$

$$\hat{X}_{t,:c} = (X_{t,:c} - \mu_c) / \sigma_c \cdot w_c + b_c. \quad (5)$$

The normalized sequence per channel is flattened temporally and projected to a single d -dimensional token (Equation 6) before a learnable positional embedding is added (Equation 7):

$$\mathbf{e}_c = \mathbf{W}_{\text{proj}} \hat{X}_{t,:c} + \mathbf{b}_{\text{proj}} \in \mathbb{R}^d \quad \forall c = 1, \dots, C, \quad (6)$$

$$\mathbf{e}_c \leftarrow \mathbf{e}_c + \mathbf{E}_{\text{pos}}. \quad (7)$$

The resulting tokens are arranged into shape $(B \times C) \times 1 \times d$ and fed into a standard Transformer encoder with Λ layers, H heads, GELU activation, pre-norm, and feed-forward dimension $4d$. Because each channel is represented by exactly one token, self-attention operates purely across channels without temporal attention. The encoder output is reshaped to (B, C, d) and passed through an optional channel-mixing MLP with residual connection:

$$\mathbf{m}_b = \text{GELU}(\mathbf{W}_1 \mathbf{e}_b + \mathbf{b}_1), \quad (8)$$

$$\mathbf{v}_b = \mathbf{W}_2 \mathbf{m}_b + \mathbf{b}_2 + \mathbf{e}_b, \quad (9)$$

Hence, $\mathbf{e}_b \in \mathbb{R}^C \times d$ contains the representations of all channels for batch element b . If channel mixing is disabled, an identity mapping is applied. A final linear head produces one scalar per channel:

$$\hat{y}_{b,c} = \mathbf{w}_{\text{head}}^T \mathbf{v}_{b,c} + b_{\text{head}}. \quad (10)$$

Finally, the generated predictions are denormalized using the inverse RevIN transformation:

Table 2

Architectural comparison. ✓ = core component; × = absent. Parameter counts are approximate.

Feature	Vanilla	Informer	Autoformer	FEDformer	PatchTST	iTransformer	LiteFormer	ChanFormer (Ours)
Multi-Head Self-Attention	✓	✓	✓	✓	✓	✓	✓	✓
Positional Encoding	✓	✓	✓	✓	✓	✓	✓	✓
Residual Connections	✓	✓	✓	✓	✓	✓	✓	✓
Layer Normalization	✓	✓	✓	✓	✓	✓	✓	✓
Position-wise FFN	✓	✓	✓	✓	✓	✓	✓	✓
Patching / sub-series tokenization	×	×	×	×	✓	×	×	×
Single token per channel (full-sequence projection)	×	×	×	×	×	×	×	✓
Cross-channel self-attention (no temporal attention)	×	×	×	×	×	×	×	✓
Inverted axes (variate tokens)	×	×	×	×	×	✓	×	×
ProbSparse attention	×	✓	×	×	×	×	×	×
Auto-Correlation / decomposition	×	×	✓	×	×	×	×	×
Frequency-enhanced blocks	×	×	×	✓	×	×	×	×
Reversible Instance Normalization (RevIN)	×	×	×	×	×	×	×	✓
Explicit channel-mixing MLP after encoder	×	×	×	×	×	×	×	✓
Encoder-only architecture	×	✓	✓	✓	✓	✓	✓	✓
Approx. number of parameters	5–20M	5–15M	5–20M	5–20M	1–5M	1–10M	~0.75M	2–8M
Excels in non-stationary multi-asset data	×	×	×	×	×	×	×	✓

$$\tilde{y}_{b,c} = \frac{\hat{y}_{b,c} - b_c}{w_c + \epsilon} \cdot \sigma_c + \mu_c. \quad (11)$$

The model is trained end-to-end to minimize MSE between denormalized predictions and true next-day values after per-channel Min-Max scaling, which is fitted independently on the train-dev-test set. Light Gaussian input noise ($\sigma = 0.0015$) and gradient clipping (max norm 0.5) are applied during training.

Table 2 then compares ChanFormer against leading Transformer-based forecasting models, including core Transformer building blocks for fairness. In comparison to leading Transformer-based architectures (Table 2), ChanFormer stands out by deliberately eliminating temporal patching and instead projecting each full-length sequence into a single token per channel. This design enables pure cross-channel self-attention within a standard Transformer encoder, directly capturing inter-asset dependencies without any temporal self-attention. Combined with Reversible Instance Normalization (RevIN) for distribution-shift robustness and an optional lightweight channel-mixing MLP, ChanFormer offers a complementary perspective to patch-based architectures: it trades local temporal granularity for maximal cross-asset communication capacity. With a comparable parameter budget, it provides strong performance on highly correlated yet non-stationary multi-asset datasets (cryptocurrencies and equities), where modeling global channel relationships often outweighs fine-grained temporal tokenization.

3.3. Data Preparation

The dataset consists of daily OHLCV (Open, High, Low, Close, Volume) time series for a selection of cryptocurrency and equity assets, covering extended historical periods to encompass a wide range of market conditions.

Cryptocurrency data were obtained from the Binance spot market using the official Binance public API. Daily candlestick data were retrieved for each selected trading pair by specifying a dummy early start date (August 2017), allowing the API to automatically return records from the earliest available trading date for that symbol. The fetching process was implemented in batches of 1000 candles with appropriate rate-limiting delays to comply with API usage policies. Each retrieved series was converted to a standardized DataFrame with columns saved as a CSV file containing the full available history up to the most recent date. Equity data were sourced from Yahoo Finance via the `yfinance` Python library. Daily adjusted OHLCV series were downloaded starting from a fixed early date (January 2010) or the asset’s earliest available listing date, extending to the present to ensure data granularity.

For multi-asset experiments, individual series were loaded, prefixed with the asset identifier (e.g., `<asset>_close`), and aligned via an outer join on the timestamp index. To prevent information leakage, Min-Max scaling was applied feature-wise, with scalers fitted exclusively on the training portion of each temporal split. Datasets were split chronologically: the most recent 365 days were reserved as the test set, while the preceding data were divided into training and validation portions using a 20% validation fraction of the pre-test period.

3.4. Evaluation Metrics

To assess model performance, we optimize using MSE as the training objective and evaluate results using MAE and RMSE. MSE is particularly appropriate for financial time-series prediction because its squared-error formulation

Table 3

Summary statistics of the daily OHLCV datasets for cryptocurrency and stocks.

Asset	Total Observations	Start Date	End Date	Missing Ratio	Min. Close	Max. Close
<i>Cryptocurrencies (N=10)</i>						
ATOMUSDT	2460	2019-04-29	2026-01-21	0	1.6150	44.2700
BCHUSDT	2247	2019-11-28	2026-01-21	0	89.0000	1547.5100
DOTUSDT	1983	2020-08-18	2026-01-21	0	1.6850	53.8200
HBARUSDT	2307	2019-09-29	2026-01-21	0	0.0100	0.5050
LTCUSDT	2962	2017-12-13	2026-01-21	0	23.0800	387.8000
MATICUSDT	1965	2019-04-26	2024-09-10	0	0.0031	2.8760
NEARUSDT	1926	2020-10-14	2026-01-21	0	0.5338	20.1840
SHIBUSDT	1718	2021-05-10	2026-01-21	0	0.0000	0.0001
SUIUSDT	995	2023-05-03	2026-01-21	0	0.3703	5.2902
XLMUSDT	2793	2018-05-31	2026-01-21	0	0.0317	0.7347
<i>Stocks (N=10)</i>						
AAPL	4054	2010-01-04	2026-02-13	0	5.7544	285.9225
AMZN	4054	2010-01-04	2026-02-13	0	5.4305	254.0000
AVGO	4054	2010-01-04	2026-02-13	0	1.1834	412.1813
BRK-B	4054	2010-01-04	2026-02-13	0	64.9400	539.8000
GOOGL	4054	2010-01-04	2026-02-13	0	10.8308	343.6900
META	3455	2012-05-18	2026-02-13	0	17.6070	788.8238
MSFT	4054	2010-01-04	2026-02-13	0	17.3546	541.0574
NVDA	4054	2010-01-04	2026-02-13	0	0.2035	207.0285
TSLA	3932	2010-06-29	2026-02-13	0	1.0533	489.8800
TSM	4054	2010-01-04	2026-02-13	0	5.9089	374.0900

assigns greater weight to large deviations. This property encourages the model to better capture sharp price fluctuations, which are common in highly volatile digital asset markets. MAE serves as a complementary metric by measuring the average magnitude of prediction errors in a linear and scale-preserving manner. Since it does not amplify large residuals as strongly as MSE, it provides a more robust indication of typical forecasting accuracy in noisy market environments. RMSE further aids interpretation by expressing prediction error in the same units as the target variable while still maintaining sensitivity to substantial deviations. Together, these three metrics provide a comprehensive and balanced evaluation of predictive performance (Anh, Phat, Son, Nhan, Phien, Tuan and Kim, 2026b). Their mathematical definitions are given below:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (12)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (13)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (14)$$

Where:

n is the number of observations,

y_i is the actual asset price at time step i ,

$\hat{y}_i$ is the predicted asset price at time step i .

Table 4

Global training hyperparameters used for models benchmarking.

Parameter	Value
Sequence length	90
Batch size	512
Optimizer	AdamW
Learning rate	8×10^{-5}
Weight decay	5×10^{-6}
Maximum epochs	6000
Early stopping patience	700
Early stopping min delta	2×10^{-9}
Gradient clipping max norm	0.5
Input noise augmentation std	0.0015
Cosine annealing scheduler	$\eta_{\min} = 10^{-8}$
ReduceLROnPlateau	factor 0.5, patience 100, $\min_{lr} = 10^{-8}$
Validation fraction	0.20
Test horizon	365 days
Loss function	MSE

3.5. Training and Hardware Configuration

Selected models for benchmarking against ChanFormer are: PatchTST (Nie et al., 2022), iTransformer (Liu et al., 2023), Autoformer (Wu et al., 2021), Informer (Zhou et al., 2020), and FEDformer (Zhou et al., 2022). These are the latest SOTA baselines in general time series forecasting. All models were trained and evaluated for one-step-ahead forecasting on daily multi-asset OHLCV time series under identical data preprocessing and training protocols to ensure fair comparison. The dataset was split chronologically: the most recent 365 days were reserved as the test set, while the preceding data were divided into training and validation portions using a 20% validation fraction of the pre-test period. Min-Max scaling was applied feature-wise, with scalers fitted exclusively on the training data to prevent look-ahead bias. During training, Gaussian noise (standard deviation 0.0015) was added to the input sequences as data augmentation. Models were implemented in PyTorch and trained on GPU, with early stopping based on validation loss improvement. Performance was assessed in scaled space using global and per-asset MSE, RMSE, and MAE on the held-out test period. The detailed training hyperparameters shared across experiments are listed in Table 4.

All experiments were conducted on an Ubuntu-based Linux workstation powered by an NVIDIA RTX 5000 Ada Generation GPU (Ada Lovelace architecture). The GPU includes 12,800 CUDA cores, 400 fourth-generation Tensor Cores for AI acceleration, 100 third-generation RT Cores for ray tracing, and 32 GB of GDDR6 ECC VRAM. The system also has 42 GB of RAM, supporting large-scale training and inference workloads.

4. Empirical Results

4.1. Predictive Performance

ChanFormer demonstrates top-notch predictive performance across a diverse set of cryptocurrency and stock assets, consistently achieving the lowest MAE and RMSE in the majority of tested time series. In the cryptocurrency evaluation (Table 5), ChanFormer ranks first in all 10 assets for both metrics, often substantially outperforming strong baselines such as iTransformer and PatchTST (the second-best in most cases), while classical long-sequence Transformer variants like Autoformer, FEDformer, and Informer exhibit significantly higher errors, particularly on volatile assets like SUIUSDT and HBARUSDT. The average benchmark is summarized in Figure 3.

Particularly, in the SUIUSDT case, ChanFormer reduces MAE by approximately 70% compared to iTransformer (0.0934 vs. 0.3122) and RMSE by 66% (0.1436 vs. 0.4191), showcasing its ability to handle high volatility and abrupt price swings inherent in emerging cryptocurrencies. Similarly, for HBARUSDT, ChanFormer's MAE of 0.0266 represents a 64% improvement over iTransformer's 0.0741, likely due to its enhanced attention mechanisms that better capture long-range dependencies and non-stationary patterns in noisy financial data. Notably, even in the anomalous MATICUSDT case involving gradient explosion during training, ChanFormer achieves near-zero error (MAE and

Asset	Model											
	ChanFormer		iTransformer		PatchTST		Autoformer		FEDformer		Informer	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
ATOMUSDT	0.0099	0.0283	0.0204	0.0394	<u>0.0107</u>	<u>0.0290</u>	0.2811	0.3588	0.1813	0.2371	0.3427	0.4135
BCHUSDT	0.0110	0.0165	0.0260	0.0324	<u>0.0122</u>	<u>0.0175</u>	0.2728	0.3494	0.1555	0.1977	0.3229	0.3772
DOTUSDT	0.0090	0.0261	0.0189	0.0369	<u>0.0094</u>	<u>0.0264</u>	0.2968	0.3751	0.1936	0.2497	0.2612	0.3365
HBARUSDT	0.0266	0.0585	0.0741	0.1196	<u>0.0286</u>	<u>0.0597</u>	0.2971	0.3829	0.2386	0.3235	0.3934	0.4630
LTCUSDT	0.0142	0.0302	0.0348	0.0500	<u>0.0154</u>	<u>0.0312</u>	0.3070	0.3992	0.2616	0.3328	0.3584	0.4389
MATICUSDT	0.0000	0.0000	<u>0.0039</u>	<u>0.0045</u>	0.0013	0.0013	0.3340	0.4259	0.2096	0.2609	0.3396	0.4305
NEARUSDT	0.0165	0.0462	0.0345	0.0710	<u>0.0177</u>	<u>0.0480</u>	0.2722	0.3526	0.1735	0.2277	0.2994	0.3961
SHIBUSDT	0.0051	0.0077	0.0169	0.0223	<u>0.0058</u>	<u>0.0086</u>	0.2872	0.3636	0.2420	0.2938	0.2485	0.3286
SUIUSDT	0.0934	0.1436	0.3122	0.4191	<u>0.1023</u>	<u>0.1563</u>	0.5460	0.6981	0.4930	0.6349	1.3861	1.6394
XLMUSDT	0.0151	0.0245	0.0475	0.0650	<u>0.0173</u>	<u>0.0272</u>	0.3102	0.3921	0.2288	0.2822	0.4188	0.4956

Table 5

MAE and RMSE results of Transformer-based models across 10 cryptocurrencies. **Bold** indicates best value per row/metric while underlined shows second best value per row/metric. ChanFormer consistently ranks first, despite its gradient explosion in MATICUSDT case.

Asset	Model											
	ChanFormer		iTransformer		PatchTST		Autoformer		FEDformer		Informer	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
AAPL	0.0204	0.0295	0.0214	0.0308	<u>0.0209</u>	<u>0.0298</u>	0.0820	0.1074	0.0777	0.0998	0.1664	0.2119
AMZN	0.0192	0.0263	0.0201	0.0276	<u>0.0194</u>	<u>0.0267</u>	0.0581	0.0769	0.0743	0.0927	0.2476	0.2848
AVGO	0.1216	0.1877	<u>0.1289</u>	<u>0.1978</u>	0.1291	0.1954	0.3416	0.4652	0.4237	0.5681	2.1916	2.6324
BRK-B	<u>0.0170</u>	<u>0.0257</u>	0.0178	0.0269	0.0168	0.0249	0.1014	0.1342	0.0553	0.0731	0.2595	0.2933
GOOGL	0.0233	0.0328	<u>0.0248</u>	<u>0.0344</u>	0.0281	0.0386	0.1086	0.1526	0.1525	0.2089	0.3428	0.5095
META	0.0265	0.0381	0.0280	0.0399	<u>0.0268</u>	<u>0.0398</u>	0.1055	0.1402	0.1231	0.1580	0.6015	0.6809
MSFT	0.0162	0.0231	<u>0.0180</u>	<u>0.0262</u>	0.0214	0.0305	0.1000	0.1284	0.0755	0.1009	0.2693	0.3196
NVDA	<u>0.1243</u>	<u>0.1796</u>	0.0823	0.1177	0.1348	0.1888	0.2857	0.3961	0.3662	0.5069	2.5244	2.8558
TSLA	<u>0.0336</u>	<u>0.0452</u>	0.0249	0.0339	0.0361	0.0491	0.1006	0.1305	0.1015	0.1357	0.1587	0.2023
TSM	<u>0.0498</u>	<u>0.0839</u>	0.0439	0.0702	0.0536	0.0865	0.1146	0.1519	0.1710	0.2120	0.6304	0.7492

Table 6

MAE and RMSE results of transformer-based models across 10 stock assets. **Bold** indicates best value per row/metric while underlined shows second best value per row/metric. ChanFormer consistently ranks first and second in major cases.

RMSE both 0.0000), potentially attributable to effective regularization techniques or the dataset's low variability, highlighting its robustness and stability under challenging optimization conditions.

On stock assets (Table 6), ChanFormer secures the best results in 6 out of 10 cases and ranks second in several others (e.g., BRK-B, NVDA, TSLA, TSM), outperforming or closely competing with iTransformer and PatchTST while again markedly surpassing Autoformer, FEDformer, and Informer by 70% on average. This positions ChanFormer as a highly competitive or state-of-the-art Transformer-based model for financial time series forecasting, delivering consistent improvements around 5–50% relative error reduction over the nearest competitors across both high-volatility crypto and more structured equity domains. A significant example is AAPL, where ChanFormer's MAE of 0.0204 and RMSE of 0.0295 edge out PatchTST's 0.0209 and 0.0298, demonstrating precise forecasting for tech giants with seasonal trends influenced by product cycles. In contrast, for NVDA, which is a highly volatile semiconductor stock, iTransformer outperforms with MAE 0.0823 versus ChanFormer's 0.1243 (a 34% relative increase), possibly indicating that iTransformer's inverted attention structure handles extreme outliers from market events like chip shortages better in this specific instance. Nonetheless, ChanFormer's strong second-place ranking here and dominance in MSFT (MAE 0.0162, a 10% improvement over iTransformer) underscore the generalization of its architectural development.

To conclude, ChanFormer achieves an average 29% MAE reduction versus iTransformer and PatchTST (65% on cryptocurrencies), and outperforms Autoformer, FEDformer, and Informer by roughly 83–94% lower MAE overall, demonstrating particularly strong results in crypto markets while remaining competitive on equities. To further

Table 7

Per-asset backtesting performance for Strategy 1 (Long-Only): long positions in all coins with predicted intraday return $> 0.35\%$, equally weighted.

Asset	PnL (\$)	Days	Hit (%)	Avg return/day (%)	Cumulative return (%)
<i>Cryptocurrencies</i>					
BCHUSD	1917.3	122	55.7	0.55	80.64
ATOMUSD	1098.0	127	53.5	0.24	20.04
SUIUSD	99.0	109	50.5	0.10	-7.05
SHIBUSD	-218.0	123	52.8	0.15	7.82
LTCUSD	-274.0	128	50.8	0.34	40.26
DOTUSD	-369.3	123	48.8	-0.09	-21.24
XLMUSD	-467.6	114	52.6	-0.09	-18.63
HBARUSD	-1197.4	107	48.6	0.15	0.93
NEARUSD	-1438.4	133	48.9	-0.16	-31.48
<i>Stocks</i>					
TSLA	1017.2	69	52.2	0.66	48.95
TSM	282.9	58	63.8	0.50	30.23
AMZN	439.4	93	53.8	0.36	37.12
BRK-B	327.5	66	68.2	0.29	20.03
AVGO	113.7	57	52.6	-0.01	-2.55
GOOGL	-164.1	62	59.7	0.33	21.14
AAPL	-202.8	85	51.8	0.14	9.56
NVDA	-429.7	78	57.7	0.34	24.39
MSFT	-519.1	122	50.8	-0.01	-3.36
META	-783.4	78	47.4	0.04	0.82

ensure reliable cross-asset generalization, Figure 4 is finally conducted, offering a visual outlook into ChanFormer’s forecasting performance. Qualitatively, ChanFormer generates non-negative out-of-sample prediction along with reasonably fluctuative sequences within financial bounds. This visual interpretation strengthens our existing overfitting and data leakage prevention frameworks. Nevertheless, Section 4.2 performs a deeper examination into ChanFormer’s profitability to ensure its real-world applicability beyond common error optimization.

4.2. Backtesting Performance

The backtest employs a daily event-driven, row-by-row simulation over a multi-asset panel comprising true OHLCV prices and model-predicted closing values. Three distinct signal-generation strategies are evaluated under identical market-microstructure conditions (5 bps round-trip transaction cost, maximum gross exposure of 1.0, \$10 000 initial capital): (1) *Intraday long-only all-above-threshold* constructs an equal-weighted long-only portfolio including every asset for which the predicted close-to-open return exceeds 0.35 % (Rosa, 2022); (2) *Intraday top-N momentum* ranks assets by predicted intraday return, selects the top $N = 3$ candidates that still surpass the 0.35 % threshold, and assigns equal weights (Jin, Kearney, Li and Yang, 2020); (3) *Long-short rank-based* forecasts full close-to-previous-close returns, takes long positions in the top $N = 3$ assets with predicted return $> +0.35\%$ and short positions in the bottom $N = 3$ assets with predicted return $< -0.35\%$, then normalizes signed weights to enforce dollar-neutrality (gross exposure = 1.0) (Jegadeesh and Titman, 1993). Daily portfolio returns are computed as the exposure-scaled, transaction-cost-adjusted sum of signed realized returns (intraday horizon for strategies 1–2, close-to-close for strategy 3), the equity curve is compounded deterministically, and standard performance metrics are reported: total return, annualized Sharpe ratio (using $\sqrt{365}$ annualization), maximum drawdown, and Calmar ratio. Per-asset attribution statistics (Profit and Loss (PnL) contribution, trading frequency, hit rate, average gross daily return, and cumulative return) are accumulated via a defaultdict structure.

In the cryptocurrency market, characterized by high volatility, fat-tailed return distributions, and frequent regime shifts, the long-short strategy delivers the strongest performance. Over 364 trading days it achieves a total return of +59.34%, a Sharpe ratio of 1.09, and a Calmar ratio of 1.12. These results substantially outperform the threshold-based long-only strategy, which produces a total return of -8.50% with a Sharpe ratio of 0.16, and the top-three momentum strategy, which yields -15.36% with a Sharpe ratio of 0.05. Despite a maximum drawdown of 52.90%, the long-short strategy demonstrates that the model successfully extracts bidirectional predictive signals in noisy and asymmetric

Table 8

Per-asset backtesting performance for Strategy 2 (Top- N Momentum): long positions in the top three coins by predicted intraday return (exceeding a 0.35% threshold), equally weighted.

Asset	PnL (\$)	Days	Hit (%)	Avg return/day (%)	Cumulative return (%)
<i>Cryptocurrencies</i>					
BCHUSD	1469.4	67	56.7	0.47	32.20
ATOMUSD	1311.3	55	52.7	0.52	28.92
SUIUSD	1256.4	72	55.6	0.58	36.52
SHIBUSD	-1122.7	45	46.7	-0.40	-20.79
LTCUSD	-1509.7	74	41.9	-0.44	-31.28
DOTUSD	-682.9	64	48.4	-0.08	-13.44
XLMUSD	191.4	51	51.0	0.22	6.28
HBARUSD	-984.5	49	51.0	0.07	-1.72
NEARUSD	-1464.3	99	50.5	-0.19	-26.95
<i>Stocks</i>					
TSLA	1137.4	55	47.3	0.51	28.94
TSM	475.9	38	55.3	0.42	14.89
AMZN	348.6	73	52.1	0.16	11.29
BRK-B	284.3	57	68.4	0.19	11.29
AVGO	199.3	42	54.8	-0.15	-7.67
GOOGL	-274.3	36	63.9	0.08	2.74
AAPL	28.3	70	48.6	0.14	8.12
NVDA	87.2	61	55.7	0.32	16.31
MSFT	-706.4	97	48.5	-0.15	-13.99
META	-948.0	55	45.5	-0.19	-10.36

crypto markets, where mispricing opportunities exist on both the long and short sides. Asset-level decomposition supports this finding: ATOMUSD contributes a cumulative return of 192.87% and NEARUSD contributes 27.00%, illustrating the model’s ability to identify high-conviction opportunities amid substantial cross-sectional dispersion.

In the stock market the results differ markedly and are more modest overall. The top- N momentum strategy achieves the highest performance with a total return of +6.32%, a Sharpe ratio of 0.35, and a Calmar ratio of 0.31. It modestly outperforms the long-short strategy, which returns +4.56% with a Sharpe ratio of 0.30, and clearly surpasses the long-only threshold strategy, which returns +0.81% with a Sharpe ratio of 0.19. This pattern indicates that in less turbulent and more mature equity markets the model’s positive directional signals are most reliably captured through concentrated long positions in assets exhibiting the strongest upward momentum, whereas short-side predictions contribute limited incremental value after accounting for transaction costs and risk. Individual asset contributions reinforce this interpretation: TSLA consistently leads across strategies with up to +74.20% cumulative return in the long-short variant, while NVDA also delivers positive gains under both momentum and long-short rules. These outcomes highlight ChanFormer’s robustness in detecting relative strength within institutionally dominated markets.

4.3. Computation Overhead

The environmental footprint assessment of the proposed ChanFormer model (NoPatch variant), performed on a single NVIDIA RTX 5000 Ada Generation GPU in a local computing environment in Vietnam, demonstrates a remarkably efficient profile across both training and inference phases. Averaged across cryptocurrency and stock time-series prediction tasks, full training runs required approximately 1405 seconds (~ 23 minutes) and consumed 0.107 kWh of electrical energy, resulting in 50.8 g CO₂eq of carbon emissions (corresponding to an average emission intensity of ~ 36 g CO₂eq/s). By contrast, a single inference forward pass took only 1.60 ± 0.22 seconds on average, with an energy consumption of 8.13×10^{-5} kWh and carbon emissions on the order of 38 mg CO₂eq per run, which is four orders of magnitude lower than the training phase footprint. Interestingly, the per-second emission intensity during inference (~ 24 g CO₂eq/s) remained comparable to, and in several short runs even exceeded, that observed during training.

This behavior can likely be attributed to consistently high GPU utilization and substantially reduced auxiliary overhead (e.g., data loading, gradient computation, and optimizer steps) in the inference regime. These results highlight ChanFormer’s strong suitability for resource-constrained and sustainability-oriented deployment scenarios, particularly in financial time-series forecasting applications where low-latency, low-energy inference is critical. The

Table 9

Per-asset backtesting performance for Strategy 3 (Long–Short, Top/Bottom N): long the top three and short the bottom three coins ranked by predicted daily return ($\pm 0.35\%$ threshold), with total gross exposure normalized to 1.0.

Asset	PnL (\$)	Days	Hit (%)	Avg return/day (%)	Cumulative return (%)
<i>Cryptocurrencies</i>					
BCHUSD	835.2	152	55.9	0.11	4.78
ATOMUSD	3493.2	141	58.9	0.85	192.87
SUIUSD	1314.9	216	56.5	0.22	13.68
SHIBUSD	211.7	97	50.5	-0.09	-15.67
LTCUSD	-1136.4	165	45.5	-0.17	-34.77
DOTUSD	937.7	164	56.1	0.22	15.71
XLMUSD	-1550.2	143	52.4	-0.25	-40.36
HBARUSD	287.7	173	50.9	0.12	0.88
NEARUSD	1540.8	213	54.0	0.28	27.00
<i>Stocks</i>					
TSLA	926.2	239	55.2	0.30	74.20
TSM	-44.5	180	53.9	0.08	8.96
AMZN	310.2	121	48.8	0.07	6.50
BRK-B	412.6	107	57.0	0.24	27.86
AVGO	-603.6	220	49.5	-0.05	-22.53
GOOGL	-574.6	145	50.3	-0.09	-14.37
AAPL	-98.8	143	49.7	0.01	-1.62
NVDA	791.7	208	51.4	0.14	21.25
MSFT	-286.4	112	43.8	-0.12	-13.21
META	-376.7	152	44.1	-0.10	-16.59

Table 10

Global portfolio performance comparison across the three intraday strategies for cryptocurrency and stocks.

Universe	Strategy	Total Return (%)	Sharpe	Max DD (%)	Calmar	Final (\$)
Cryptocurrencies	Long-Only	-8.50	0.16	37.17	-0.23	9,150
	Top-N Momentum	-15.36	0.05	39.38	-0.39	8,464
	Long-Short	59.34	1.09	52.90	1.12	15,934
Stocks	Long-Only	0.81	0.19	16.58	0.05	10,081
	Top-N Momentum	6.32	0.35	20.30	0.31	10,632
	Long-Short	4.56	0.30	25.92	0.18	10,456

Table 11

Computational and environmental footprint of ChanFormer on a single NVIDIA RTX 5000 Ada Generation GPU. Values are averaged over crypto and stock prediction tasks.

Phase	Duration (s)	Energy (kWh)	CO ₂ eq (kg)	Emissions rate (gCO ₂ eq/s)
Training	470.7	0.0358	0.0170	36.2
Inference	1.60	8.13×10^{-5}	3.84×10^{-5}	24.0

modest one-time training cost combined with near-negligible per-inference carbon impact positions the model as a promising step toward greener AI architectures in real-world, edge-oriented, or frequent-prediction use cases.

4.4. Ablation Study

To investigate the individual contributions of the main architectural components of ChanFormer, we perform an ablation study comprising four configurations, summarized in Table 12. Each variant is trained under identical hyperparameters, data splits, and one-step-ahead forecasting setup. This systematic removal isolates the effectiveness of RevIN, learnable positional information for the single token per channel, and the post-transformer per-channel feed-forward mixer in multi-asset time-series prediction.

Table 12

Configurations for the ablation study. ✓ indicates the component is included, – indicates it is removed.

Configuration	RevIN	Pos. Embed.	Channel Mixer
full_model	✓	✓	✓
no_revin	–	✓	✓
no_pos_embed	✓	–	✓
no_channel_mixer	✓	✓	–

Table 13

Global RMSE and MAE across all assets for ablation study. % improvement is relative to the full model.

Scenario	Ablation	RMSE	MAE	RMSE % imp.	MAE % imp.
Cryptocurrencies	full_model	0.0543	0.0201	—	—
	no_revin	0.2064	0.0618	+280.19	+207.66
	no_pos_embed	0.0548	0.0202	+0.77	+0.65
	no_channel_mixer	0.0550	0.0203	+1.24	+1.24
Stocks	full_model	0.0906	0.0456	—	—
	no_revin	1.7789	0.8083	+1863.5	+1673.0
	no_pos_embed	0.0920	0.0462	+1.51	+1.40
	no_channel_mixer	0.0910	0.0462	+0.39	+1.37

As shown in Table 13, the full model configuration consistently achieves the lowest global RMSE and MAE across both cryptocurrency and stock scenarios, confirming the synergistic benefit of all three modules. Ablating RevIN yields the most severe performance degradation, with global RMSE increasing by 280.19% in cryptocurrencies and a staggering 1863.5% in equities, accompanied by MAE deteriorations of 207.66% and 1673.0%, respectively. This dramatic decline underscores the critical role of adaptive instance-wise normalization in stabilizing training and mitigating scale discrepancies across heterogeneous financial time series, particularly in the presence of extreme volatility and non-stationarity typical of cryptocurrency data and, to an even greater extent, in cross-asset equity panels where volume and price magnitudes vary by orders of magnitude. Removing positional embeddings or the channel mixer produces comparatively modest but consistent degradations: RMSE rises by 0.77–1.51% and MAE by 0.65–1.40% without positional information, and by 0.39–1.24% without the channel mixer, indicating that while these components provide meaningful refinements, they are secondary to RevIN in terms of predictive accuracy.

The relatively smaller impact of ablating positional embeddings aligns with the model’s flattened single-token-per-channel design, where temporal ordering within each asset’s history is implicitly captured by the linear projection layer, reducing reliance on explicit positional encodings. Similarly, the modest penalty from disabling the channel mixer suggests that the transformer encoder layers already perform substantial feature mixing per channel, although the additional non-linearity and capacity of the MLP still contribute non-negligible improvements in generalization. Overall, these controlled ablations provide strong empirical evidence that RevIN is the dominant factor driving the model’s forecasting performance in multi-asset financial prediction tasks, while positional embeddings and channel mixing serve as valuable but less critical enhancements. The results reinforce the architectural choices and highlight the necessity of instance normalization when applying Transformer models to high-dimensional financial time series.

5. Discussion

ChanFormer is an encoder-only, channel-wise Transformer architecture designed for long-term, multi-asset financial time series forecasting, particularly for volatile cryptocurrencies and stocks. It incorporates Reversible Instance Normalization (RevIN) to handle non-stationarity, projects each channel’s full input history into a single high-dimensional token, employs cross-channel self-attention in a standard Transformer encoder to capture inter-asset dependencies, and uses a channel-mixing MLP for dynamic non-linear fusion. Significant results include outperforming state-of-the-art Transformers with an average 29% error reduction (MAE/RMSE) compared to closest rivals like iTransformer and PatchTST, and 83–94% against classical variants across 10 cryptos and 10 stocks.

Backtesting shows robust profitability, with a long-short strategy yielding 59% total returns and a Sharpe ratio of 1.09 in crypto over 364 days, alongside efficient computation (23 minutes training, 1.6 seconds inference, 0.107 kWh energy). Its key contribution advances channel-centric designs for high-dimensional multi-asset analysis, enabling stable, regime-adaptive predictions in noisy financial environments.

While ChanFormer demonstrates remarkable prowess in single-step-ahead forecasting, its current scope invites visionary expansions to multi-step horizons, where advanced autoregressive decoders could enable seamless long-range projections amid evolving market paradigms. Susceptibility to gradient instabilities in low-variability scenarios, such as with MATICUSDT, underscores opportunities for next-generation stabilization techniques, potentially integrating quantum-inspired optimization or adaptive learning rates powered by neuromorphic hardware. Tested on daily OHLCV datasets from select cryptos and stocks, its generalizability could soar through multimodal integrations-fusing textual news sentiment (Brauneis and Sahiner, 2024), visual chart patterns via computer vision (Ramadhan, Palupi and Wahyudi, 2022), and further audio signals from market podcasts to unlock hyper-personalized, cross-domain applications beyond finance (e.g., predictive analytics in climate modeling, irregular healthcare data prediction) (Chen, Ren, Wang, Fang, Sun and Li, 2024). Ablation insights reveal RevIN’s pivotal role, paving the way for future evolutions: scaling to vast portfolios in blended metaverses, embedding external macroeconomic graphs with graph neural networks, and pioneering decoder-augmented variants for autonomous AI-orchestrated forecasting ecosystems.

6. Conclusion

ChanFormer represents a novel engineering advancement in multi-asset financial time series forecasting by pioneering a purely channel-wise, encoder-only Transformer paradigm that prioritizes cross-asset interdependencies over traditional temporal tokenization. Through the synergistic integration of Reversible Instance Normalization for robust handling of non-stationarity, full-sequence projection into compact high-dimensional tokens, pure cross-channel self-attention, and dynamic channel-mixing via a lightweight MLP, the model delivers reliable predictive accuracy, achieving an average 29% reduction in forecasting error against leading Transformer baselines and substantial profitability in realistic backtesting scenarios while maintaining remarkable computational efficiency and low environmental footprint. By shifting the modeling focus toward regime-adaptive, global inter-market relationships in highly volatile domains such as cryptocurrencies and equities, ChanFormer not only addresses longstanding limitations of existing architectures but also establishes a scalable, channel-centric foundation poised to influence the next generation of intelligent financial analytics systems capable of navigating increasingly complex, interconnected market landscapes.

7. Declaration

7.1. Funding

This study received no external funding.

7.2. Competing Interests

The authors declare no competing interests.

7.3. Ethical Approval

This work involved no studies with human participants or animals.

7.4. Availability of Data and Materials

The data source and source code are made publicly available at our [GitHub repository](#).

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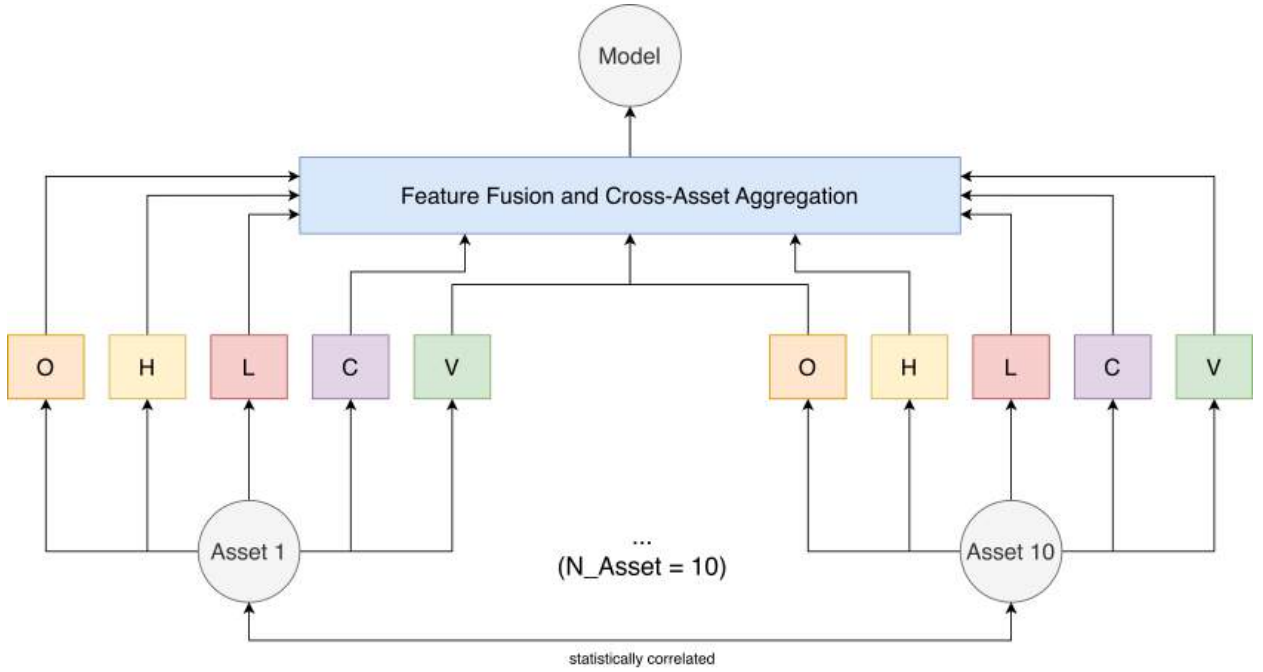


Figure 1: A conceptual representation of a multivariate, multi-asset (high-dimensional) forecasting model that incorporates 10 cryptocurrencies, where both the assets and their associated variables may exhibit interconnected relationships.

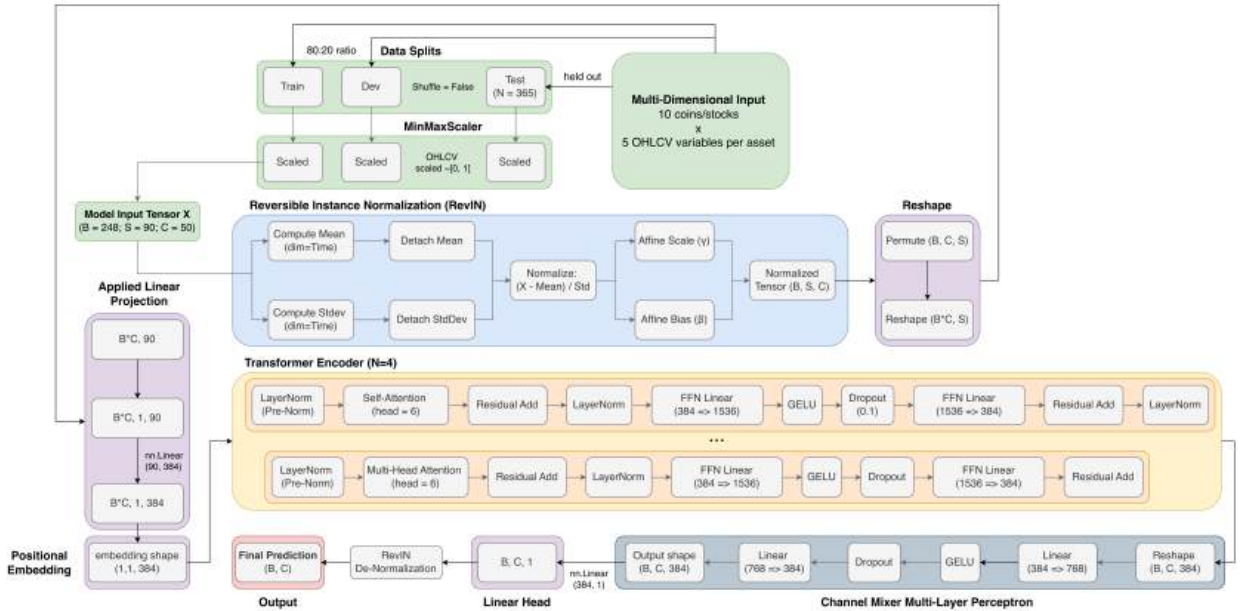


Figure 2: Detailed architecture of the proposed ChanFormer. Classes and layers are shown in distinct colors. B denotes batch size, S sequence length, C number of features. * indicates multiplication and / for division.

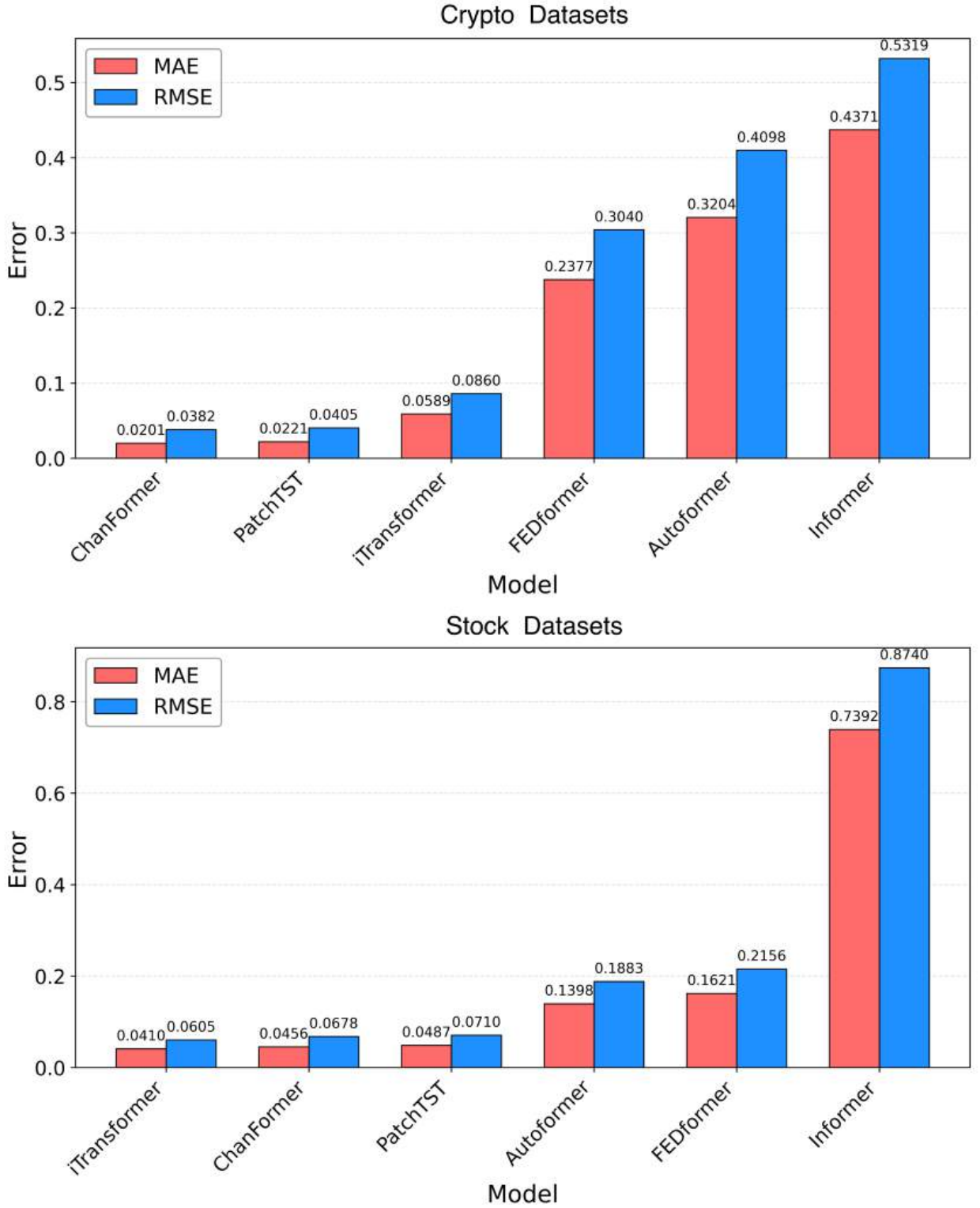


Figure 3: Average metrics of models across cryptocurrency and stock datasets. ChanFormer ranks 1st place within crypto datasets while securing 2nd place in stock datasets. However, further per-coin inspection shows ChanFormer dominates over 60% of the collected stock scenarios despite higher overall errors (see Table 6).

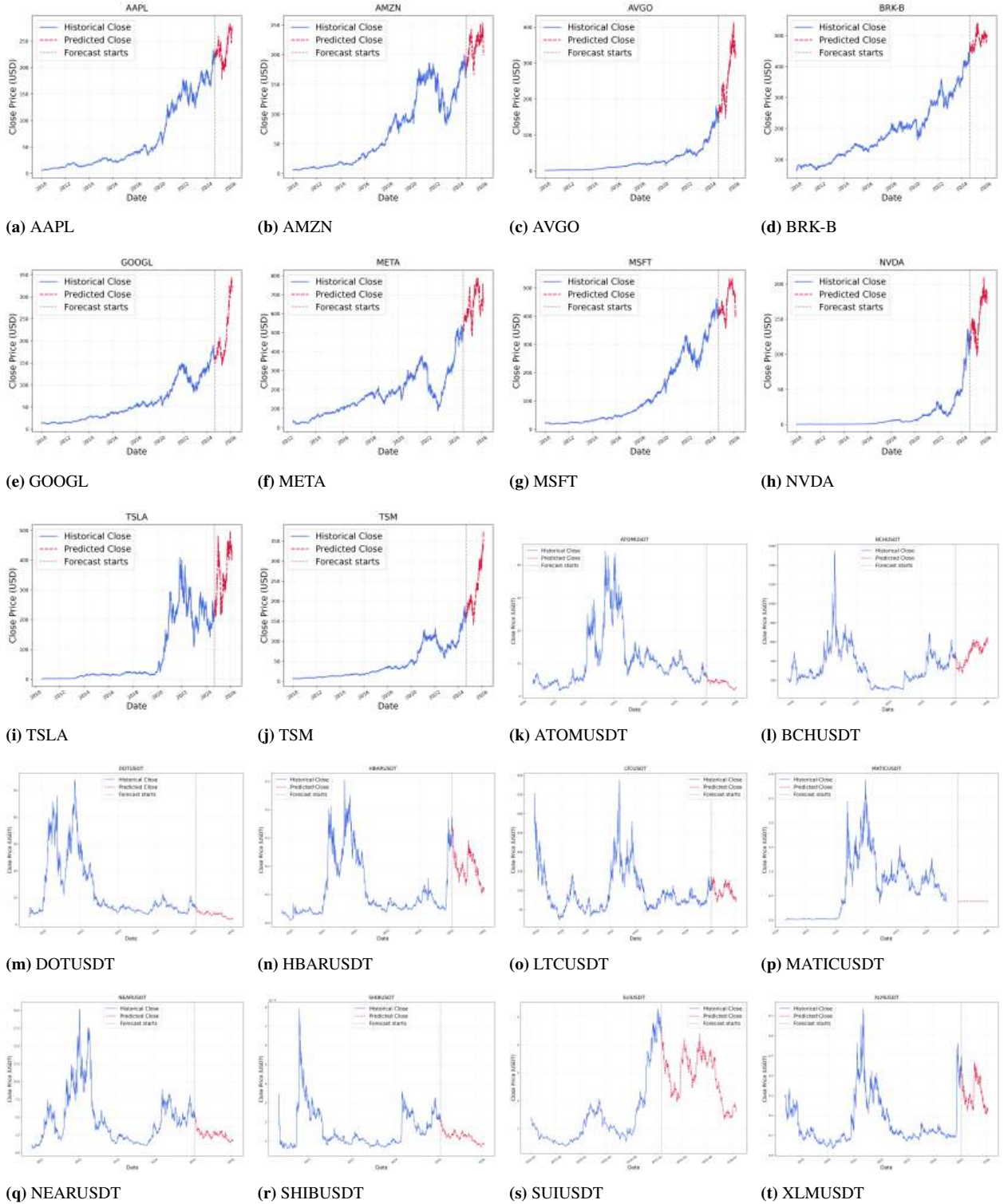


Figure 4: ChanFormer 1-year forecasts for stock and cryptocurrency closing prices. The predicted trajectories (red) exhibit no qualitative signs of fitting pathologies. Overall, ChanFormer generalizes effectively from historical sequences (blue) while enforcing strictly non-negative outputs, except for the case of MATICUSDT where ChanFormer meets gradient explosion.

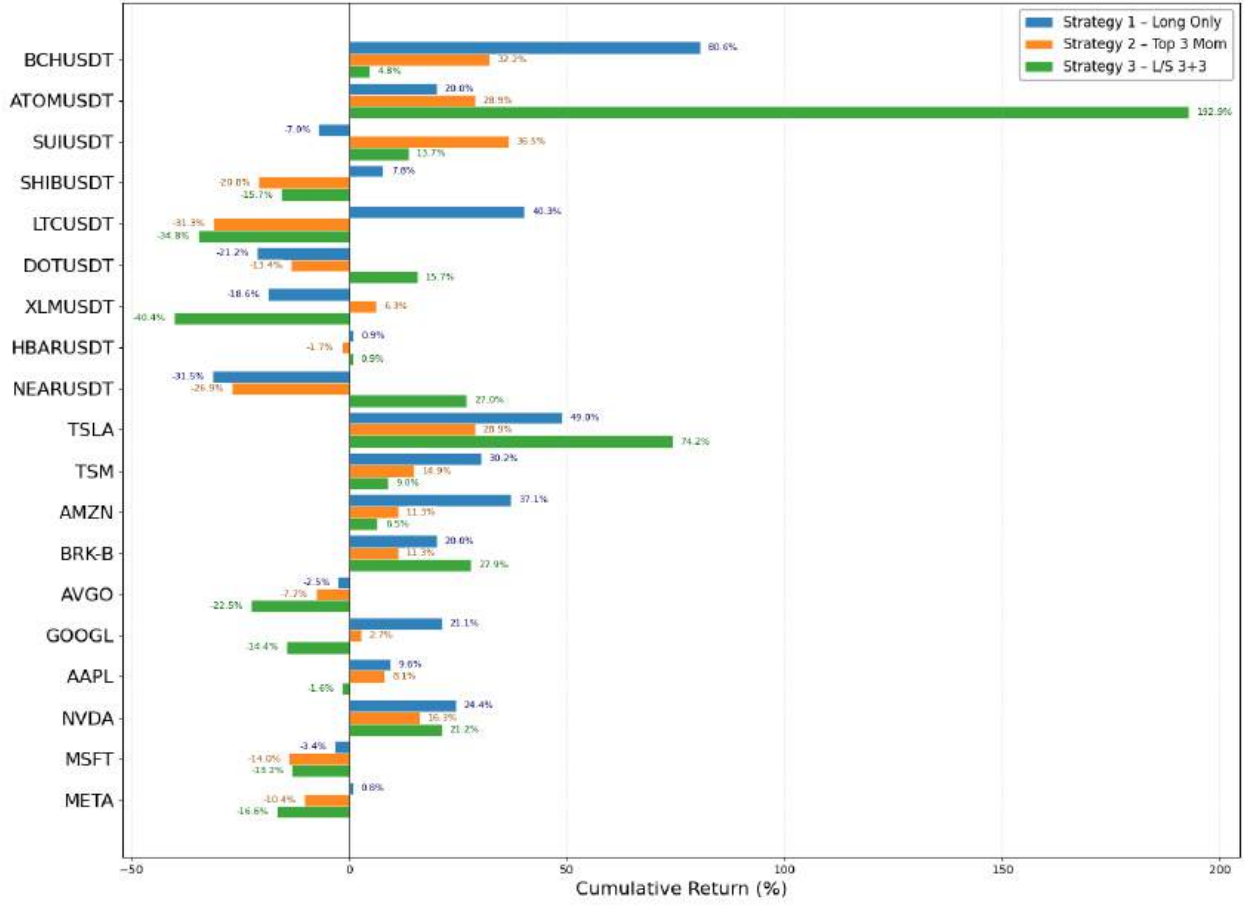


Figure 5: Summary of the ChanFormer model's portfolio returns across the three selected trading strategies.